

Design and implement a relational database for a **University Management System** using SQL Data Definition Language (DDL) commands. The goal is to construct the database schema, define appropriate column data types, establish primary and foreign key constraints, and enforce data integrity rules.

Create a database named `UniversityDB` consisting of the following 10 entities:

Departments (department_id [PK], department_name, office_location, phone, email)

Programs (program_id [PK], program_name, degree, duration, department_id [FK])

Students (student_id [PK], first_name, last_name, gender, date_of_birth, email, phone, address, admission_date, program_id [FK])

Faculty (faculty_id [PK], first_name, last_name, designation, email, phone, department_id [FK])

Courses (course_id [PK], course_code, course_title, credit, semester, department_id [FK])

Classrooms (classroom_id [PK], room_number, building_name, capacity)

Sections (section_id [PK], section_name, semester, year, course_id [FK], faculty_id [FK], classroom_id [FK])

Enrollments (enrollment_id [PK], student_id [FK], section_id [FK], enrollment_date, grade)

Examinations (exam_id [PK], section_id [FK], exam_type, exam_date, total_marks)

Results (result_id [PK], exam_id [FK], student_id [FK], obtained_marks, grade)

- Write the SQL statement to create `UniversityDB` and set it as the active database context prior to creating tables.
- Assign optimal standard SQL data types (e.g., `INT`, `VARCHAR`, `DATE`, `DECIMAL`) for each attribute across all 10 entities.
- Explicitly declare Primary Keys (**PK**) for all tables and define Foreign Keys (**FK**) to establish relational integrity between tables.
- Sequence your `CREATE TABLE` statements logically so that parent tables are created before any child tables referencing them with foreign keys.
- Apply `NOT NULL` constraints to essential fields such as names, dates, codes, and foreign key columns.
- Apply `UNIQUE` constraints to all `email` attributes and `course_code`.
- Apply `CHECK` constraints where applicable (e.g., course `credit` > 0, classroom `capacity` > 0).

- Save all SQL statements in a single script file named `[RollNumber]_Project1.sql` .
- Include a header comment at the beginning of the SQL file containing your Full Name, Roll Number, and Submission Date.
- Ensure that the entire SQL script executes sequentially without runtime or foreign key constraint errors.